

Algorithms & Complexity

03

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Interview-ready reference: first principles, Python snippets, comparison tables, and cheat sheets.

1 Overview

Algorithmic thinking is the foundation of every coding interview. You are not just expected to solve problems — you are expected to **reason about efficiency**, identify optimal approaches, and communicate trade-offs clearly. This file covers the complete algorithmic toolkit needed for top-tier software engineering interviews.

What interviewers evaluate:

1. Can you identify what kind of problem this is? (sorting, graph, DP, greedy...)
2. Do you know the right algorithm and its complexity?
3. Can you implement it correctly under pressure?
4. Can you articulate trade-offs and edge cases?

Core skill loop: Recognize → Design → Analyze → Implement → Verify → Optimize

2 Complexity Analysis

2.1 Big-O / Big-Theta / Big-Omega

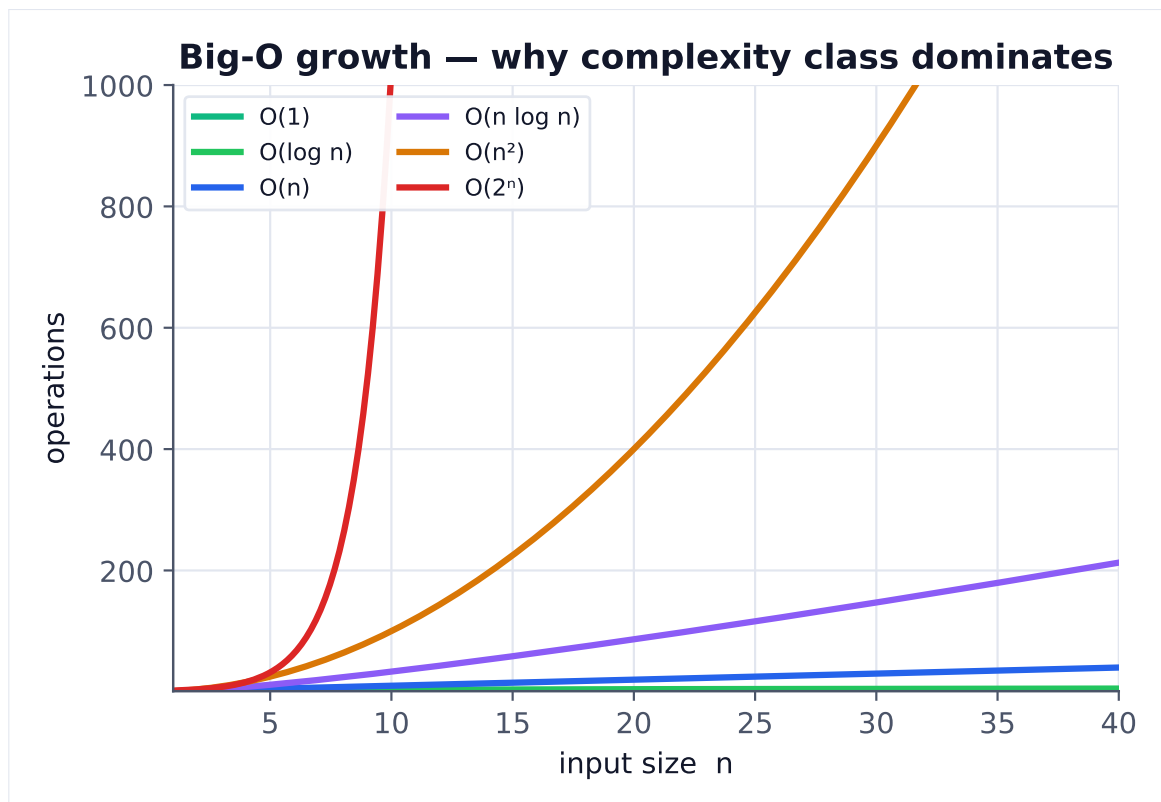


Figure 1. Asymptotic class, not constant factors, decides scalability. Beyond modest n , an $O(n \log n)$ algorithm crushes $O(n^2)$; exponential is infeasible past $n=30$.

Complexity notation describes **how resource usage scales with input size**, not actual runtime.

Notation	Name	Meaning
$O(f(n))$	Big-O (upper bound)	Algorithm runs in at most $f(n)$ steps (worst case)
$\Omega(f(n))$	Big-Omega (lower bound)	Algorithm runs in at least $f(n)$ steps (best case)
$\Theta(f(n))$	Big-Theta (tight bound)	Algorithm runs in exactly $f(n)$ steps (avg case matches both)

Formal definition – Big-O: $T(n) = O(f(n))$ if $\exists$ constants $c > 0$ and n_0 such that $T(n) \leq c \cdot f(n)$ for all $n \geq n_0$.

Complexity hierarchy (slowest to fastest growth):

$O(1) < O(\log n) < O(\sqrt{n}) < O(n) < O(n \log n) < O(n^2) < O(n^3) < O(2^n) < O(n!)$

Dropping constants and lower-order terms:

- › $O(3n^2 + 5n + 12) \rightarrow O(n^2)$
- › $O(100) \rightarrow O(1)$
- › $O(n + n \log n) \rightarrow O(n \log n)$

Multi-variable complexity: When input has multiple dimensions, keep all:

- › $O(V + E)$ for graph algorithms (V = vertices, E = edges)
- › $O(m \cdot n)$ for 2D DP problems

2.2 Amortized Analysis

Amortized analysis gives the **average cost per operation over a sequence**, even if individual operations are expensive.

Example – Dynamic Array (Python list append):

- › Most appends: $O(1)$
- › Occasional doubling: $O(n)$
- › Amortized cost per append: $O(1)$ – because doubling happens exponentially rarely

Three methods: | Method | Idea | When to use | |-----|-----|-----| | Aggregate | Total cost / n operations | Simplest; works when all operations have same amortized cost | | Accounting | Prepay credits for future expensive ops | When operations have different costs | | Potential | $\Phi(\text{state})$ function tracks "stored energy" | Most general; used for complex data structures |

Key amortized results:

- › Dynamic array append: $O(1)$ amortized
- › Union-Find with union-by-rank + path compression: $O(\alpha(n))$ amortized $\approx O(1)$
- › Splay tree operations: $O(\log n)$ amortized
- › Fibonacci heap decrease-key: $O(1)$ amortized

2.3 Computing Time & Space Complexity

Time complexity rules:

1. **Sequential statements:** Add costs – $O(A) + O(B) = O(\max(A, B))$
2. **Nested loops:** Multiply – outer $\times$ inner
3. **Function calls:** Substitute the called function's complexity
4. **Recursion:** Use recurrence relations or Master Theorem

```

1 # O(n) – single loop
2 for i in range(n):
3     do_constant_work()
4
5 # O(n²) – nested loops
6 for i in range(n):
7     for j in range(n):
8         do_constant_work()
9
10 # O(n log n) – loop + halving inner
11 for i in range(n):
12     j = n
13     while j > 1:
14         j /= 2 # log n iterations
15
16 # O(log n) – binary search
17 lo, hi = 0, n
18 while lo < hi:
19     mid = (lo + hi) // 2
20     # halves search space each time

```

Space complexity – what to count:

- › Input space (usually excluded unless asked for auxiliary space)
- › Call stack depth for recursion
- › Extra data structures allocated
- › **Auxiliary space** = space excluding input

```

1 # O(1) space – no extra allocation
2 def find_max(arr):
3     m = arr[0]
4     for x in arr:
5         m = max(m, x)
6     return m
7
8 # O(n) space – stores extra array
9 def prefix_sum(arr):
10    ps = [0] * (len(arr) + 1) # O(n)
11    for i, x in enumerate(arr):
12        ps[i+1] = ps[i] + x
13    return ps
14
15 # O(n) space – call stack for recursion
16 def factorial(n):
17     if n ≤ 1: return 1
18     return n * factorial(n - 1) # n frames on stack

```

2.4 Recursion Tree & Master Theorem

Recursion tree method: Draw the tree of recursive calls, compute cost at each level, sum all levels.

$$T(n) = 2T(n/2) + n \quad (\text{merge sort})$$

Level 0: n → cost n
 Level 1: $n/2 + n/2$ → cost n
 Level 2: $n/4 + n/4 + n/4 + n/4$ → cost n

...
 Level $\log n$: n leaves of size 1 $\rightarrow$ cost n
 Total: $n \times (\log n + 1) = O(n \log n)$

Master Theorem: For recurrences of the form $T(n) = a \cdot T(n/b) + f(n)$ where $a \geq 1$, $b > 1$:
 Let critical exponent: $c = \log_b(a)^*$

Case	Condition	Result	Intuition
Case 1	$f(n) = O(n^{(c^* - \varepsilon)})$ for some $\varepsilon > 0$	$T(n) = \Theta(n^{c^*})$	Leaves dominate
Case 2	$f(n) = \Theta(n^{c^*} \cdot \log^k n)$, $k \geq 0$	$T(n) = \Theta(n^{c^*} \cdot \log^{(k+1)} n)$	Equal work per level
Case 3	$f(n) = \Omega(n^{(c^* + \varepsilon)})$ and regularity holds	$T(n) = \Theta(f(n))$	Root dominates

Master Theorem examples:

Recurrence	a	b	c^*	$f(n)$	Case	Result
$T(n) = 2T(n/2) + n$	2	2	1	$n = \Theta(n^1)$	2	$\Theta(n \log n)$
$T(n) = 2T(n/2) + 1$	2	2	1	$1 = O(n^{0.5})$	1	$\Theta(n)$
$T(n) = 4T(n/2) + n^2$	4	2	2	$n^2 = \Theta(n^2)$	2	$\Theta(n^2 \log n)$
$T(n) = 4T(n/2) + n^3$	4	2	2	$n^3 = \Omega(n^{2.5})$	3	$\Theta(n^3)$
$T(n) = T(n/2) + 1$	1	2	0	$1 = \Theta(1)$	2	$\Theta(\log n)$
$T(n) = 3T(n/3) + n$	3	3	1	$n = \Theta(n^1)$	2	$\Theta(n \log n)$

Master Theorem does NOT apply when:

- ▶ Not in form $a \cdot T(n/b) + f(n)$ — e.g., $T(n) = T(n-1) + n$ (linear decrease, not fraction)
- ▶ $T(n) = T(n/2) + T(n/3) + n$ (unequal subproblems)
- ▶ For $T(n) = T(n-1) + n \rightarrow \text{sum } 1+2+\dots+n = O(n^2)$
- ▶ For $T(n) = T(n-1) + 1 \rightarrow n \text{ recursive calls} \rightarrow O(n)$

2.5 Input Size $\rightarrow$ Acceptable Complexity

Use this table to instantly identify which algorithms are feasible given constraints.

n (input size)	Max acceptable complexity	Typical algorithm
$n \leq 10$	$O(n!)$ or $O(n^n)$	Brute-force, permutations
$n \leq 20$	$O(2^n \cdot n)$	Bitmask DP, meet in the middle
$n \leq 100$	$O(n^3)$ or $O(n^4)$	Floyd-Warshall, naive string matching
$n \leq 1,000$	$O(n^2)$	Insertion sort, naive DP
$n \leq 10,000$	$O(n^2 \log n)$	Slower $O(n^2)$ algorithms
$n \leq 100,000$	$O(n \log n)$ or $O(n\sqrt{n})$	Merge sort, segment tree, sqrt decomp
$n \leq 1,000,000$	$O(n \log n)$	Merge sort, heap operations
$n \leq 10,000,000$	$O(n)$	Linear scans, counting sort
$n > 10^8$	$O(\log n)$ or $O(1)$	Binary search, math formulas

Rule of thumb: Modern computers do $\sim 10^8$ simple operations/second. Design accordingly.

3 Algorithms

3.1 Sorting

Sorting algorithms at a glance					
	Best	Average	Worst	Space	Stable
Bubble	$O(n \log n)$	$O(n^2)$	$O(n^2)$	$O(1)$	yes
Insertion	$O(n \log n)$	$O(n^2)$	$O(n^2)$	$O(1)$	yes
Selection	$O(n^2)$	$O(n^2)$	$O(n^2)$	$O(1)$	no
Merge	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$	yes
Quick	$O(n \log n)$	$O(n \log n)$	$O(n^2)$	$O(\log n)$	no
Heap	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(1)$	no
Counting	$O(n)$	$O(n)$	$O(n)$	$O(n)$	yes
Radix	$O(n)$	$O(n)$	$O(n)$	$O(n)$	yes

Figure 2. No single sort wins everywhere: merge sort is stable and worst-case $O(n \log n)$ but needs $O(n)$ space; quicksort is faster in practice but $O(n^2)$ worst case.

Comparison-Based Sorts

Comparison sorts have a theoretical lower bound of $\Omega(n \log n)$ — you cannot do better with only comparison operations.

Merge Sort

```

1 def merge_sort(arr):
2     if len(arr) ≤ 1:
3         return arr
4     mid = len(arr) // 2
5     left = merge_sort(arr[:mid])
6     right = merge_sort(arr[mid:])
7     return merge(left, right)
8
9 def merge(left, right):
10    result = []
11    i = j = 0
12    while i < len(left) and j < len(right):
13        if left[i] ≤ right[j]: # ≤ preserves stability
14            result.append(left[i]); i += 1
15        else:
16            result.append(right[j]); j += 1
17    result.extend(left[i:])
18    result.extend(right[j:])
19    return result
  
```

Key insight: Divide conquer and merge. Recurrence $T(n) = 2T(n/2) + n \rightarrow O(n \log n)$ always.
Stability: Yes — equal elements keep original relative order. **Drawback:** $O(n)$ extra space; cache-unfriendly for large arrays.

Quick Sort

```

1 import random
2
3 def quicksort(arr, lo, hi):
4     if lo < hi:
5         p = partition(arr, lo, hi)
6         quicksort(arr, lo, p - 1)
7         quicksort(arr, p + 1, hi)
8
9 def partition(arr, lo, hi):
10     # Lomuto partition scheme
11     # Randomize pivot to avoid worst-case  $O(n^2)$  on sorted input
12     rand_idx = random.randint(lo, hi)
13     arr[rand_idx], arr[hi] = arr[hi], arr[rand_idx]
14     pivot = arr[hi]
15     i = lo - 1
16     for j in range(lo, hi):
17         if arr[j] ≤ pivot:
18             i += 1
19             arr[i], arr[j] = arr[j], arr[i]
20     arr[i + 1], arr[hi] = arr[hi], arr[i + 1]
21     return i + 1
22
23 # Hoare partition (more efficient in practice):
24 def hoare_partition(arr, lo, hi):
25     pivot = arr[(lo + hi) // 2]
26     i, j = lo - 1, hi + 1
27     while True:
28         i += 1
29         while arr[i] < pivot: i += 1
30         j -= 1
31         while arr[j] > pivot: j -= 1
32         if i ≥ j: return j
33         arr[i], arr[j] = arr[j], arr[i]

```

Key insight: Pick pivot, partition so $\text{left} \leq \text{pivot} \leq \text{right}$, recurse on both sides. **Pivot strategies:**

- › Last element: $O(n^2)$ on sorted/reverse-sorted input
- › Random pivot: Expected $O(n \log n)$, avoids adversarial inputs
- › Median-of-three: Good heuristic
- › **3-way partition (Dutch National Flag):** Best for arrays with many duplicates — partitions into $< \text{pivot}, = \text{pivot}, > \text{pivot}$

Stability: No — partition swaps break relative order.

Heap Sort

```

1 def heap_sort(arr):
2     n = len(arr)
3     # Build max-heap (heapify from last internal node)
4     for i in range(n // 2 - 1, -1, -1):
5         heapify(arr, n, i)

```

```

6     # Extract elements
7     for i in range(n - 1, 0, -1):
8         arr[0], arr[i] = arr[i], arr[0] # move max to end
9         heapify(arr, i, 0)
10
11 def heapify(arr, n, i):
12     largest = i
13     left, right = 2*i + 1, 2*i + 2
14     if left < n and arr[left] > arr[largest]: largest = left
15     if right < n and arr[right] > arr[largest]: largest = right
16     if largest != i:
17         arr[i], arr[largest] = arr[largest], arr[i]
18         heapify(arr, n, largest)

```

Key insight: Build max-heap in $O(n)$, then repeatedly extract-max in $O(\log n)$ each. **Stability:** No. **Space:** $O(1)$ in-place (advantage over merge sort).

Insertion Sort

```

1 def insertion_sort(arr):
2     for i in range(1, len(arr)):
3         key = arr[i]
4         j = i - 1
5         while j ≥ 0 and arr[j] > key:
6             arr[j + 1] = arr[j]
7             j -= 1
8         arr[j + 1] = key
9     return arr

```

Key insight: Build sorted portion left-to-right by inserting each element in its correct position. **Best case:** $O(n)$ when already sorted (inner loop never executes). **When to use:** Small arrays ($n < 20$), nearly-sorted data, online algorithms (elements arrive one by one). **Stability:** Yes.

Non-Comparison Sorts

Non-comparison sorts break the $O(n \log n)$ barrier by exploiting structure of keys.

Counting Sort

```

1 def counting_sort(arr, max_val):
2     count = [0] * (max_val + 1)
3     for x in arr:
4         count[x] += 1
5     # Build cumulative counts for stability
6     for i in range(1, max_val + 1):
7         count[i] += count[i - 1]
8     output = [0] * len(arr)
9     for x in reversed(arr): # reverse for stability
10        output[count[x] - 1] = x
11        count[x] -= 1
12    return output

```

Complexity: $O(n + k)$ time, $O(n + k)$ space where k = range of values. **When to use:** Keys are integers in a known small range. Useless if $k \gg n$.

Radix Sort

```

1 def radix_sort(arr):
2     if not arr: return arr
3     max_val = max(arr)
4     exp = 1
5     while max_val // exp > 0:
6         arr = counting_sort_by_digit(arr, exp)
7         exp *= 10
8     return arr
9
10 def counting_sort_by_digit(arr, exp):
11     n = len(arr)
12     output = [0] * n
13     count = [0] * 10
14     for x in arr:
15         count[(x // exp) % 10] += 1
16     for i in range(1, 10):
17         count[i] += count[i - 1]
18     for x in reversed(arr):
19         idx = (x // exp) % 10
20         output[count[idx] - 1] = x
21         count[idx] -= 1
22     return output

```

Complexity: $O(d \cdot (n + k))$ where d = digits, k = base (10 for decimal). Effectively $O(n)$ for fixed-width integers. **Stability:** Requires stable inner sort (counting sort). Must process digits from **least significant** to most significant (LSD radix sort).

Bucket Sort

```

1 def bucket_sort(arr, num_buckets=10):
2     if not arr: return arr
3     min_val, max_val = min(arr), max(arr)
4     bucket_range = (max_val - min_val) / num_buckets + 1e-9
5     buckets = [[] for _ in range(num_buckets)]
6     for x in arr:
7         idx = int((x - min_val) / bucket_range)
8         idx = min(idx, num_buckets - 1)
9         buckets[idx].append(x)
10    result = []
11    for b in buckets:
12        b.sort()      # insertion sort works well for small buckets
13        result.extend(b)
14    return result

```

Complexity: $O(n + k)$ average when data is uniformly distributed; $O(n^2)$ worst case if all elements land in one bucket. **When to use:** Floating-point numbers in a known range, uniformly distributed data.

Sorting Comparison Table

Algorithm	Best	Average	Worst	Space	Stable	Notes
Merge Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$	Yes	Guaranteed; good for linked lists
Quick Sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$	$O(\log n)^*$	No	Fastest in practice; randomize pivot

Algorithm	Best	Average	Worst	Space	Stable	Notes
Heap Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(1)$	No	Cache-unfriendly
Insertion Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	Yes	Best for small/nearly-sorted
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$	$O(1)$	No	Never use in interviews
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	Yes	Never use in interviews
Counting Sort	$O(n+k)$	$O(n+k)$	$O(n+k)$	$O(n+k)$	Yes	Only integers in small range
Radix Sort	$O(d \cdot n)$	$O(d \cdot n)$	$O(d \cdot n)$	$O(n+k)$	Yes	Fixed-width keys
Bucket Sort	$O(n+k)$	$O(n+k)$	$O(n^2)$	$O(n+k)$	Yes	Uniform float data
Tim Sort	$O(n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$	Yes	Python's built-in; hybrid merge+insertion

*Quick sort call stack: $O(\log n)$ average, $O(n)$ worst.

Stability matters when: You sort by multiple keys sequentially, or the original order of equal elements has meaning (e.g., sort students by grade then by name).

When to use which:

- › **General purpose:** Quick sort (in-place, fast cache) or merge sort (stable)
- › **Guaranteed worst case:** Merge sort or heap sort
- › **Memory-constrained:** Heap sort ($O(1)$ space)
- › **Nearly sorted:** Insertion sort or Tim sort
- › **Integer keys, small range:** Counting sort
- › **Large integers, many of same size:** Radix sort
- › **Floating point, uniform:** Bucket sort
- › **Linked lists:** Merge sort (no random access needed)

3.2 Searching

Linear Search

```

1 def linear_search(arr, target):
2     for i, x in enumerate(arr):
3         if x == target:
4             return i
5     return -1

```

$O(n)$ time, $O(1)$ space. Use only when array is unsorted or n is tiny.

Binary Search

Standard binary search:

```

1 def binary_search(arr, target):
2     lo, hi = 0, len(arr) - 1
3     while lo <= hi:
4         mid = lo + (hi - lo) // 2 # avoids integer overflow
5         if arr[mid] == target:
6             return mid

```

```

7         elif arr[mid] < target:
8             lo = mid + 1
9         else:
10            hi = mid - 1
11    return -1

```

First occurrence of target:

```

1 def first_occurrence(arr, target):
2     lo, hi = 0, len(arr) - 1
3     result = -1
4     while lo ≤ hi:
5         mid = (lo + hi) // 2
6         if arr[mid] == target:
7             result = mid
8             hi = mid - 1 # keep searching left
9         elif arr[mid] < target:
10            lo = mid + 1
11        else:
12            hi = mid - 1
13    return result

```

Last occurrence of target:

```

1 def last_occurrence(arr, target):
2     lo, hi = 0, len(arr) - 1
3     result = -1
4     while lo ≤ hi:
5         mid = (lo + hi) // 2
6         if arr[mid] == target:
7             result = mid
8             lo = mid + 1 # keep searching right
9         elif arr[mid] < target:
10            lo = mid + 1
11        else:
12            hi = mid - 1
13    return result

```

Lower bound (first index where arr[i] ≥ target):

```

1 import bisect
2
3 def lower_bound(arr, target):
4     return bisect.bisect_left(arr, target)
5
6 # Manual implementation:
7 def lower_bound_manual(arr, target):
8     lo, hi = 0, len(arr) # hi = len(arr), not len-1
9     while lo < hi: # lo < hi, not lo ≤ hi
10        mid = (lo + hi) // 2
11        if arr[mid] < target:
12            lo = mid + 1

```

```

13         else:
14             hi = mid          # not mid-1
15     return lo

```

Binary search on rotated sorted array:

```

1 def search_rotated(arr, target):
2     lo, hi = 0, len(arr) - 1
3     while lo ≤ hi:
4         mid = (lo + hi) // 2
5         if arr[mid] == target:
6             return mid
7         # Left half is sorted
8         if arr[lo] ≤ arr[mid]:
9             if arr[lo] ≤ target < arr[mid]:
10                hi = mid - 1
11            else:
12                lo = mid + 1
13        # Right half is sorted
14        else:
15            if arr[mid] < target ≤ arr[hi]:
16                lo = mid + 1
17            else:
18                hi = mid - 1
19    return -1

```

Binary search on answer space (parametric search):

The most powerful binary search pattern: instead of searching an array, search for the **answer itself**.

```

1 # Example: Koko eating bananas — find minimum eating speed
2 def min_eating_speed(piles, h):
3     def can_finish(speed):
4         return sum((p + speed - 1) // speed for p in piles) ≤ h
5
6     lo, hi = 1, max(piles)
7     while lo < hi:
8         mid = (lo + hi) // 2
9         if can_finish(mid):
10            hi = mid          # mid works, try smaller
11        else:
12            lo = mid + 1      # too slow, need faster
13    return lo
14
15 # Template: "find minimum X such that condition(X) is True"
16 # lo = min possible, hi = max possible
17 # condition is monotone: False...False...True...True

```

Binary search complexity: $O(\log n)$ time, $O(1)$ space.

Binary search mental model: Maintain an invariant — the answer is always in $[lo, hi]$. Shrink the range by half each iteration. When $lo == hi$, that's the answer.

3.3 Graph Algorithms

Graphs: $G = (V, E)$ where V = vertices (nodes), E = edges.

Representations:

- › **Adjacency list:** $\text{graph}[u] = [(v, \text{weight}), \dots]$ — $O(V + E)$ space, efficient for sparse graphs
- › **Adjacency matrix:** $\text{graph}[u][v] = \text{weight}$ — $O(V^2)$ space, $O(1)$ edge lookup, efficient for dense graphs

BFS (Breadth-First Search)

```

1 from collections import deque
2
3 def bfs(graph, start):
4     visited = set([start])
5     queue = deque([start])
6     order = []
7     while queue:
8         node = queue.popleft()
9         order.append(node)
10        for neighbor in graph[node]:
11            if neighbor not in visited:
12                visited.add(neighbor)
13                queue.append(neighbor)
14    return order
15
16 # BFS shortest path (unweighted graph):
17 def bfs_shortest_path(graph, start, end):
18     if start == end: return [start]
19     visited = {start: None} # node → parent
20     queue = deque([start])
21     while queue:
22         node = queue.popleft()
23         for neighbor in graph[node]:
24             if neighbor not in visited:
25                 visited[neighbor] = node
26                 if neighbor == end:
27                     path = []
28                     cur = end
29                     while cur is not None:
30                         path.append(cur)
31                         cur = visited[cur]
32                     return path[::-1]
33                 queue.append(neighbor)
34    return [] # no path

```

Complexity: $O(V + E)$ **Use when:** Shortest path in unweighted graph, level-order traversal, connected components, bipartite check. **Key property:** Visits nodes in order of distance from source — guarantees shortest path in unweighted graphs.

DFS (Depth-First Search)

```

1 def dfs_iterative(graph, start):
2     visited = set()
3     stack = [start]
4     order = []
5     while stack:
6         node = stack.pop()

```

```

7     if node in visited:
8         continue
9     visited.add(node)
10    order.append(node)
11    for neighbor in graph[node]:
12        if neighbor not in visited:
13            stack.append(neighbor)
14    return order
15
16 def dfs_recursive(graph, node, visited=None):
17     if visited is None: visited = set()
18     visited.add(node)
19     for neighbor in graph[node]:
20         if neighbor not in visited:
21             dfs_recursive(graph, neighbor, visited)
22     return visited

```

Complexity: $O(V + E)$ **Use when:** Cycle detection, topological sort, connected components, path existence, flood fill. **Key property:** Explores as deep as possible before backtracking.

Topological Sort

Valid only for **Directed Acyclic Graphs (DAGs)**.

Kahn's Algorithm (BFS-based):

```

1  from collections import deque
2
3  def topological_sort_kahn(graph, n):
4      """
5      graph: adjacency list {node: [neighbors]}
6      n: number of nodes (0 to n-1)
7      Returns topological order, or empty list if cycle exists.
8      """
9      in_degree = [0] * n
10     for u in range(n):
11         for v in graph[u]:
12             in_degree[v] += 1
13
14     queue = deque(u for u in range(n) if in_degree[u] == 0)
15     order = []
16
17     while queue:
18         u = queue.popleft()
19         order.append(u)
20         for v in graph[u]:
21             in_degree[v] -= 1
22             if in_degree[v] == 0:
23                 queue.append(v)
24
25     if len(order) != n:
26         return [] # cycle detected - not a DAG
27     return order

```

DFS-based topological sort:

```

1 def topological_sort_dfs(graph, n):
2     WHITE, GRAY, BLACK = 0, 1, 2
3     color = [WHITE] * n
4     order = []
5     has_cycle = [False]
6
7     def dfs(u):
8         if has_cycle[0]: return
9         color[u] = GRAY
10        for v in graph[u]:
11            if color[v] == GRAY:
12                has_cycle[0] = True
13                return
14            if color[v] == WHITE:
15                dfs(v)
16        color[u] = BLACK
17        order.append(u) # add to order AFTER all descendants
18
19    for u in range(n):
20        if color[u] == WHITE:
21            dfs(u)
22
23    if has_cycle[0]:
24        return []
25    return order[::-1] # reverse post-order = topological order

```

Complexity: $O(V + E)$ for both methods. **Kahn vs DFS:**

- Kahn: Intuitive, easy cycle detection ($\text{len}(\text{order}) \neq n$), produces lexicographically smallest order with a min-heap
- DFS: More natural for recursive thinking, easier to add finish-time ordering

Use cases: Build systems (Makefile), course prerequisites, task scheduling, dependency resolution.

Dijkstra's Algorithm

Shortest path from single source; **requires non-negative edge weights.**

```

1 import heapq
2
3 def dijkstra(graph, start, n):
4     """
5     graph: adjacency list {u: [(v, weight), ...]}
6     Returns dist[] where dist[v] = shortest distance from start to v.
7     """
8     dist = [float('inf')] * n
9     dist[start] = 0
10    # Min-heap: (distance, node)
11    pq = [(0, start)]
12
13    while pq:
14        d, u = heapq.heappop(pq)
15        if d > dist[u]:
16            continue # stale entry - already found shorter path
17        for v, w in graph[u]:
18            if dist[u] + w < dist[v]:
19                dist[v] = dist[u] + w

```

```

20         heapq.heappush(pq, (dist[v], v))
21
22     return dist
23
24 # With path reconstruction:
25 def dijkstra_with_path(graph, start, end, n):
26     dist = [float('inf')] * n
27     prev = [-1] * n
28     dist[start] = 0
29     pq = [(0, start)]
30     while pq:
31         d, u = heapq.heappop(pq)
32         if d > dist[u]: continue
33         if u == end: break
34         for v, w in graph[u]:
35             if dist[u] + w < dist[v]:
36                 dist[v] = dist[u] + w
37                 prev[v] = u
38                 heapq.heappush(pq, (dist[v], v))
39
40     # Reconstruct path
41     path = []
42     cur = end
43     while cur != -1:
44         path.append(cur)
45         cur = prev[cur]
46     return dist[end], path[::-1]

```

Complexity: $O((V + E) \log V)$ with binary heap, $O(V^2 + E)$ with array (dense graphs). **When to use:** Single-source shortest path, non-negative weights. **Does NOT work with negative edges** — use Bellman-Ford instead.

Bellman-Ford Algorithm

Shortest path from single source; **handles negative edge weights**; detects negative cycles.

```

1 def bellman_ford(edges, n, start):
2     """
3     edges: list of (u, v, weight)
4     Returns dist[] or None if negative cycle reachable from start.
5     """
6     dist = [float('inf')] * n
7     dist[start] = 0
8
9     # Relax all edges V-1 times
10    for _ in range(n - 1):
11        updated = False
12        for u, v, w in edges:
13            if dist[u] != float('inf') and dist[u] + w < dist[v]:
14                dist[v] = dist[u] + w
15                updated = True
16        if not updated:
17            break # converged early
18
19    # Check for negative cycles (V-th relaxation)
20    for u, v, w in edges:
21        if dist[u] != float('inf') and dist[u] + w < dist[v]:
22            return None # negative cycle detected
23

```

```
24     return dist
```

Complexity: $O(V \cdot E)$ **When to use:** Negative edge weights, detecting negative cycles (e.g., currency arbitrage). **Guaranteed correct after $V-1$ iterations** because any shortest path visits at most $V-1$ edges (in a graph with no negative cycles).

Floyd-Warshall Algorithm

All-pairs shortest paths; handles negative edges; detects negative cycles.

```
1 def floyd_warshall(n, edges):
2     """Returns dist[i][j] = shortest path from i to j."""
3     INF = float('inf')
4     dist = [[INF] * n for _ in range(n)]
5     for i in range(n):
6         dist[i][i] = 0
7     for u, v, w in edges:
8         dist[u][v] = min(dist[u][v], w) # handle multi-edges
9
10    for k in range(n): # intermediate vertex
11        for i in range(n):
12            for j in range(n):
13                if dist[i][k] + dist[k][j] < dist[i][j]:
14                    dist[i][j] = dist[i][k] + dist[k][j]
15
16    # Check negative cycles: dist[i][i] < 0 for some i
17    for i in range(n):
18        if dist[i][i] < 0:
19            return None # negative cycle
20
21    return dist
```

Complexity: $O(V^3)$ time, $O(V^2)$ space. **When to use:** All-pairs shortest paths, transitive closure, detecting negative cycles.

Graph Algorithm Comparison Table

Algorithm	Handles Negative Edges	Handles Negative Cycles	Single/All Pairs	Complexity	Notes
Dijkstra	No	No	Single source	$O((V+E) \log V)$	Fastest for non-negative; use with priority queue
Bellman-Ford	Yes	Detects	Single source	$O(V \cdot E)$	Use when negative edges exist
Floyd-Warshall	Yes	Detects	All pairs	$O(V^3)$	Best for dense graphs, all-pairs
BFS	No (unit weights only)	No	Single source	$O(V+E)$	Optimal for unweighted graphs
DAG Shortest Path	Yes (DAG only)	N/A	Single source	$O(V+E)$	Topo sort + relax; fastest for DAGs

Minimum Spanning Tree (MST)

MST of a connected weighted undirected graph: a spanning tree with minimum total edge weight.

Kruskal's Algorithm:

```
1 def kruskal(n, edges):
2     """
3     edges: list of (weight, u, v)
4     Returns MST weight and edges.
5     """
6     edges.sort() # sort by weight
7     parent = list(range(n))
8     rank = [0] * n
9
10    def find(x):
11        while parent[x] != x:
12            parent[x] = parent[parent[x]] # path compression (2-step)
13            x = parent[x]
14        return x
15
16    def union(x, y):
17        px, py = find(x), find(y)
18        if px == py: return False # same component - cycle
19        if rank[px] < rank[py]:
20            px, py = py, px
21        parent[py] = px
22        if rank[px] == rank[py]:
23            rank[px] += 1
24        return True
25
26    mst_weight = 0
27    mst_edges = []
28    for w, u, v in edges:
29        if union(u, v):
30            mst_weight += w
31            mst_edges.append((u, v, w))
32            if len(mst_edges) == n - 1:
33                break
34
35    return mst_weight, mst_edges
```

Prim's Algorithm:

```
1 import heapq
2
3 def prim(graph, n):
4     """
5     graph: adjacency list {u: [(v, weight), ...]}
6     Returns MST weight.
7     """
8     visited = set()
9     # Start from node 0: (weight, node)
10    pq = [(0, 0)]
11    mst_weight = 0
12
13    while pq and len(visited) < n:
14        w, u = heapq.heappop(pq)
```

```

15     if u in visited:
16         continue
17     visited.add(u)
18     mst_weight += w
19     for v, weight in graph[u]:
20         if v not in visited:
21             heapq.heappush(pq, (weight, v))
22
23     return mst_weight if len(visited) == n else -1 # -1 if disconnected

```

Kruskal vs Prim: | | Kruskal | Prim | |-----|-----| | Approach | Edge-centric: sort all edges, add if no cycle | Vertex-centric: grow tree from one vertex | | Complexity | $O(E \log E)$ | $O((V+E) \log V)$ with heap | | Best for | Sparse graphs | Dense graphs | | Data structure | Union-Find | Priority queue |

Union-Find (Disjoint Set Union)

```

1 class UnionFind:
2     def __init__(self, n):
3         self.parent = list(range(n))
4         self.rank = [0] * n
5         self.components = n
6
7     def find(self, x):
8         if self.parent[x] != x:
9             self.parent[x] = self.find(self.parent[x]) # path compression
10        return self.parent[x]
11
12    def union(self, x, y):
13        px, py = self.find(x), self.find(y)
14        if px == py: return False
15        # Union by rank
16        if self.rank[px] < self.rank[py]:
17            px, py = py, px
18        self.parent[py] = px
19        if self.rank[px] == self.rank[py]:
20            self.rank[px] += 1
21        self.components -= 1
22        return True
23
24    def connected(self, x, y):
25        return self.find(x) == self.find(y)

```

Amortized complexity: $O(\alpha(n))$ per operation where α is inverse Ackermann — effectively $O(1)$. **Use for:** Cycle detection, Kruskal's MST, connected components, dynamic connectivity.

Cycle Detection

Undirected graph — Union-Find:

```

1 def has_cycle_undirected(n, edges):
2     uf = UnionFind(n)
3     for u, v in edges:
4         if not uf.union(u, v):
5             return True # u and v already in same component

```

```
6     return False
```

Undirected graph – DFS (coloring):

```
1 def has_cycle_undirected_dfs(graph, n):
2     visited = set()
3     def dfs(node, parent):
4         visited.add(node)
5         for neighbor in graph[node]:
6             if neighbor not in visited:
7                 if dfs(neighbor, node): return True
8             elif neighbor != parent: # back edge to non-parent = cycle
9                 return True
10        return False
11    for i in range(n):
12        if i not in visited:
13            if dfs(i, -1): return True
14    return False
```

Directed graph – DFS with 3-color:

```
1 def has_cycle_directed(graph, n):
2     WHITE, GRAY, BLACK = 0, 1, 2
3     color = [WHITE] * n
4     def dfs(u):
5         color[u] = GRAY
6         for v in graph[u]:
7             if color[v] == GRAY: return True # back edge = cycle
8             if color[v] == WHITE and dfs(v): return True
9         color[u] = BLACK
10        return False
11    return any(dfs(u) for u in range(n) if color[u] == WHITE)
```

Key difference: In undirected graphs, visiting a neighbor (other than parent) that's already visited = cycle. In directed graphs, visiting a GRAY (currently-being-processed) node = cycle (back edge).

Connected Components

```
1 def connected_components(graph, n):
2     visited = set()
3     components = []
4     def dfs(node, component):
5         visited.add(node)
6         component.append(node)
7         for neighbor in graph[node]:
8             if neighbor not in visited:
9                 dfs(neighbor, component)
10    for i in range(n):
11        if i not in visited:
12            component = []
13            dfs(i, component)
```

```

14         components.append(component)
15     return components

```

Bipartite Check

```

1  from collections import deque
2
3  def is_bipartite(graph, n):
4      color = [-1] * n
5      for start in range(n):
6          if color[start] != -1: continue
7          color[start] = 0
8          queue = deque([start])
9          while queue:
10             u = queue.popleft()
11             for v in graph[u]:
12                 if color[v] == -1:
13                     color[v] = 1 - color[u]
14                     queue.append(v)
15                 elif color[v] == color[u]:
16                     return False # same color = odd cycle = not bipartite
17     return True

```

A graph is bipartite iff it has no odd-length cycles. Two-color with BFS; if conflict found, not bipartite.

3.4 Dynamic Programming

DP solves problems by **breaking them into overlapping subproblems** and storing results to avoid recomputation.

Memoization vs Tabulation

	Memoization (Top-Down)	Tabulation (Bottom-Up)
Approach	Recursive with cache	Iterative, fill table
Direction	Problem → subproblems	Base cases → problem
Space	O(depth) stack + memo	O(states) table
Complexity	Same asymptotically	Often faster constants
Ease	Easier to implement	Better for optimization
When to use	Not all subproblems needed	All subproblems computed

```

1  # Memoization (Fibonacci):
2  from functools import lru_cache
3
4  @lru_cache(maxsize=None)
5  def fib_memo(n):
6      if n <= 1: return n
7      return fib_memo(n-1) + fib_memo(n-2)
8
9  # Tabulation (Fibonacci):
10 def fib_tab(n):

```

```

11     if n ≤ 1: return n
12     dp = [0, 1]
13     for i in range(2, n+1):
14         dp.append(dp[-1] + dp[-2])
15     return dp[n]
16
17 # Space-optimized (only need last 2):
18 def fib_opt(n):
19     a, b = 0, 1
20     for _ in range(n):
21         a, b = b, a + b
22     return a

```

How to Identify DP Problems

DP applies when:

1. Problem asks for **optimal value** (min/max) or **count of ways**
2. Subproblems **overlap** (same subproblem solved repeatedly)
3. Problem has **optimal substructure** (optimal solution uses optimal solutions to subproblems)
4. Problem asks: "is it possible to achieve X?"

DP does NOT apply when:

- › No overlapping subproblems (use divide and conquer)
- › Greedy works (exchange argument)
- › Subproblems are independent

DP design recipe:

1. Define the **state** — what information uniquely identifies a subproblem?
2. Write the **recurrence** (transition function)
3. Identify **base cases**
4. Determine **computation order** (which states depend on which)
5. Answer — which state(s) give the final answer?

Classic DP Families

0/1 Knapsack: Items with weight and value; capacity W ; each item used at most once. Recurrence: $dp[i][w] = \max(dp[i-1][w], dp[i-1][w - \text{weight}[i]] + \text{value}[i])$

```

1 def knapsack_01(weights, values, W):
2     n = len(weights)
3     dp = [[0] * (W + 1) for _ in range(n + 1)]
4     for i in range(1, n + 1):
5         for w in range(W + 1):
6             dp[i][w] = dp[i-1][w] # don't take item i
7             if weights[i-1] ≤ w:
8                 dp[i][w] = max(dp[i][w], dp[i-1][w - weights[i-1]] + values[i-1])
9     return dp[n][W]
10
11 # Space-optimized (1D DP) — iterate w in reverse:
12 def knapsack_01_opt(weights, values, W):
13     dp = [0] * (W + 1)
14     for i in range(len(weights)):
15         for w in range(W, weights[i] - 1, -1): # MUST go right to left

```

```

16         dp[w] = max(dp[w], dp[w - weights[i]] + values[i])
17     return dp[W]

```

Unbounded Knapsack (items can be reused):

```

1 def knapsack_unbounded(weights, values, W):
2     dp = [0] * (W + 1)
3     for w in range(W + 1):
4         for i in range(len(weights)):
5             if weights[i] ≤ w:
6                 dp[w] = max(dp[w], dp[w - weights[i]] + values[i])
7     return dp[W]

```

Coin Change (minimum coins): Recurrence: $dp[\text{amount}] = \min(dp[\text{amount} - \text{coin}] + 1)$ for each coin

```

1 def coin_change(coins, amount):
2     dp = [float('inf')] * (amount + 1)
3     dp[0] = 0
4     for a in range(1, amount + 1):
5         for coin in coins:
6             if coin ≤ a:
7                 dp[a] = min(dp[a], dp[a - coin] + 1)
8     return dp[amount] if dp[amount] ≠ float('inf') else -1

```

Coin Change (number of ways):

```

1 def coin_change_ways(coins, amount):
2     dp = [0] * (amount + 1)
3     dp[0] = 1
4     for coin in coins:           # order matters: outer=coins for combinations
5         for a in range(coin, amount + 1):
6             dp[a] += dp[a - coin]
7     return dp[amount]

```

Longest Common Subsequence (LCS): Recurrence:

- If $s1[i] == s2[j]$: $dp[i][j] = dp[i-1][j-1] + 1$
- Else: $dp[i][j] = \max(dp[i-1][j], dp[i][j-1])$

```

1 def lcs(s1, s2):
2     m, n = len(s1), len(s2)
3     dp = [[0] * (n + 1) for _ in range(m + 1)]
4     for i in range(1, m + 1):
5         for j in range(1, n + 1):
6             if s1[i-1] == s2[j-1]:
7                 dp[i][j] = dp[i-1][j-1] + 1
8             else:
9                 dp[i][j] = max(dp[i-1][j], dp[i][j-1])

```

```
10     return dp[m][n]
```

Edit Distance (Levenshtein): Operations: insert, delete, replace (each cost 1). Recurrence:

- If $s1[i] == s2[j]$: $dp[i][j] = dp[i-1][j-1]$
- Else: $dp[i][j] = 1 + \min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1])$

```

1 def edit_distance(s1, s2):
2     m, n = len(s1), len(s2)
3     dp = [[0] * (n + 1) for _ in range(m + 1)]
4     for i in range(m + 1): dp[i][0] = i
5     for j in range(n + 1): dp[0][j] = j
6     for i in range(1, m + 1):
7         for j in range(1, n + 1):
8             if s1[i-1] == s2[j-1]:
9                 dp[i][j] = dp[i-1][j-1]
10            else:
11                dp[i][j] = 1 + min(dp[i-1][j],      # delete from s1
12                                   dp[i][j-1],      # insert into s1
13                                   dp[i-1][j-1])      # replace
14     return dp[m][n]
```

Longest Increasing Subsequence (LIS):

```

1 # O(n^2) DP:
2 def lis_n2(arr):
3     n = len(arr)
4     dp = [1] * n
5     for i in range(1, n):
6         for j in range(i):
7             if arr[j] < arr[i]:
8                 dp[i] = max(dp[i], dp[j] + 1)
9     return max(dp)
10
11 # O(n log n) patience sorting:
12 import bisect
13
14 def lis_nlogn(arr):
15     tails = [] # tails[i] = smallest tail element of all IS of length i+1
16     for x in arr:
17         pos = bisect.bisect_left(tails, x)
18         if pos == len(tails):
19             tails.append(x)
20         else:
21             tails[pos] = x
22     return len(tails)
```

Matrix Chain Multiplication (Interval DP): Recurrence: $dp[i][j] = \min \text{ over } k \text{ of } (dp[i][k] + dp[k+1][j] + \text{dims}[i-1] * \text{dims}[k] * \text{dims}[j])$

```

1 def matrix_chain(dims):
2     n = len(dims) - 1 # number of matrices
3     dp = [[0] * n for _ in range(n)]
4     for length in range(2, n + 1): # subproblem length
5         for i in range(n - length + 1):
6             j = i + length - 1
7             dp[i][j] = float('inf')
8             for k in range(i, j):
9                 cost = dp[i][k] + dp[k+1][j] + dims[i] * dims[k+1] * dims[j+1]
10                dp[i][j] = min(dp[i][j], cost)
11     return dp[0][n-1]

```

Matrix Paths (unique paths, min cost path):

```

1 def unique_paths(m, n):
2     dp = [[1] * n for _ in range(m)]
3     for i in range(1, m):
4         for j in range(1, n):
5             dp[i][j] = dp[i-1][j] + dp[i][j-1]
6     return dp[m-1][n-1]
7
8 def min_path_sum(grid):
9     m, n = len(grid), len(grid[0])
10    dp = [[0] * n for _ in range(m)]
11    dp[0][0] = grid[0][0]
12    for i in range(1, m): dp[i][0] = dp[i-1][0] + grid[i][0]
13    for j in range(1, n): dp[0][j] = dp[0][j-1] + grid[0][j]
14    for i in range(1, m):
15        for j in range(1, n):
16            dp[i][j] = grid[i][j] + min(dp[i-1][j], dp[i][j-1])
17    return dp[m-1][n-1]

```

DP on Trees:

```

1 def tree_dp_max_independent_set(adj, n):
2     # dp[node][0] = max IS if node NOT included
3     # dp[node][1] = max IS if node included
4     dp = [[0, 1] for _ in range(n)] # [0] = not taken, [1] = taken (value 1)
5
6     def dfs(u, parent):
7         for v in adj[u]:
8             if v == parent: continue
9             dfs(v, u)
10            # If u not taken, v can be taken or not
11            dp[u][0] += max(dp[v][0], dp[v][1])
12            # If u taken, v cannot be taken
13            dp[u][1] += dp[v][0]
14
15    dfs(0, -1)
16    return max(dp[0][0], dp[0][1])

```

Bitmask DP (Travelling Salesman): State: $dp[mask][i]$ = minimum cost to visit all cities in mask, ending at city i .

```

1 def tsp(dist, n):
2     INF = float('inf')
3     dp = [[INF] * n for _ in range(1 << n)]
4     dp[1][0] = 0 # visited city 0, at city 0
5
6     for mask in range(1, 1 << n):
7         for u in range(n):
8             if not (mask >> u & 1): continue
9             if dp[mask][u] == INF: continue
10            for v in range(n):
11                if mask >> v & 1: continue # already visited
12                new_mask = mask | (1 << v)
13                dp[new_mask][v] = min(dp[new_mask][v], dp[mask][u] + dist[u][v])
14
15     full = (1 << n) - 1
16     return min(dp[full][v] + dist[v][0] for v in range(n))

```

DP Key Takeaways:

- › **State is everything** — spend most time defining what uniquely describes a subproblem
- › **Recurrence is mechanical** — once state is defined, transitions follow naturally
- › **Base cases are critical** — wrong base cases cause wrong answers even with correct recurrence
- › **Space optimization:** 2D DP can often become 1D; 1D can become O(1) if only adjacent states needed

3.5 Greedy Algorithms

Greedy algorithms make the **locally optimal choice at each step** hoping to reach a global optimum.

When Greedy Works

Greedy is correct when two conditions hold:

1. **Greedy choice property:** A globally optimal solution can always be reached by making locally optimal choices
2. **Optimal substructure:** Optimal solution contains optimal solutions to subproblems

Exchange argument: Standard proof technique. Assume greedy is not optimal. Show that any optimal solution can be transformed (by swapping) to match greedy without increasing cost — contradiction.

Classic Greedy Problems

Activity Selection (Interval Scheduling): Select maximum number of non-overlapping intervals. Strategy: Sort by end time; always pick the activity that finishes earliest.

```

1 def activity_selection(intervals):
2     intervals.sort(key=lambda x: x[1]) # sort by end time
3     count = 0
4     last_end = float('-inf')
5     for start, end in intervals:
6         if start >= last_end:
7             count += 1
8             last_end = end
9     return count

```

Huffman Coding: Build optimal prefix-free codes using a min-heap. Greedy: always merge the two lowest-frequency nodes.

Fractional Knapsack: (unlike 0/1 knapsack, greedy works here)

```

1 def fractional_knapsack(weights, values, W):
2     ratios = sorted(zip(values, weights), key=lambda x: x[0]/x[1], reverse=True)
3     total = 0
4     for v, w in ratios:
5         if W >= w:
6             total += v; W -= w
7         else:
8             total += v * W / w; break
9     return total

```

Jump Game: Can you reach the end of an array where each element is max jump length?

```

1 def can_jump(nums):
2     max_reach = 0
3     for i, jump in enumerate(nums):
4         if i > max_reach: return False
5         max_reach = max(max_reach, i + jump)
6     return True

```

Task Scheduling: Schedule tasks with deadlines to maximize profit. Strategy: Sort by profit descending; greedily assign tasks to latest available slot.

Gas Station: Find starting station to complete circular tour.

```

1 def can_complete_circuit(gas, cost):
2     total_surplus = 0
3     surplus = 0
4     start = 0
5     for i in range(len(gas)):
6         diff = gas[i] - cost[i]
7         total_surplus += diff
8         surplus += diff
9         if surplus < 0:
10            start = i + 1
11            surplus = 0
12    return start if total_surplus >= 0 else -1

```

Greedy vs DP: Greedy is $O(n \log n)$ or $O(n)$; DP is $O(n^2)$ or worse. **Prefer greedy when provably correct** (exchange argument or known greedy structure). DP is the safe fallback.

3.6 Recursion & Backtracking

Recursion Template

```

1 def solve(state, ...):
2     # Base case
3     if is_complete(state):
4         record_solution(state)

```

```

5         return
6
7     # Recursive case
8     for choice in get_choices(state):
9         if is_valid(choice, state):
10             make_choice(state, choice)
11             solve(state, ...)
12             undo_choice(state, choice) # backtrack

```

Backtracking Framework

Backtracking = DFS on the decision tree with **pruning** to avoid exploring invalid paths.

```

1  # Subsets (power set):
2  def subsets(nums):
3      result = []
4      def backtrack(start, path):
5          result.append(path[:])
6          for i in range(start, len(nums)):
7              path.append(nums[i])
8              backtrack(i + 1, path)
9              path.pop()
10     backtrack(0, [])
11     return result
12
13 # Permutations:
14 def permutations(nums):
15     result = []
16     def backtrack(path, used):
17         if len(path) == len(nums):
18             result.append(path[:])
19             return
20         for i in range(len(nums)):
21             if used[i]: continue
22             used[i] = True
23             path.append(nums[i])
24             backtrack(path, used)
25             path.pop()
26             used[i] = False
27     backtrack([], [False] * len(nums))
28     return result
29
30 # N-Queens:
31 def solve_n_queens(n):
32     result = []
33     cols = set(); diag1 = set(); diag2 = set()
34     board = [['.' for _ in range(n)] for _ in range(n)]
35     def backtrack(row):
36         if row == n:
37             result.append(''.join(r) for r in board)
38             return
39         for col in range(n):
40             if col in cols or (row-col) in diag1 or (row+col) in diag2:
41                 continue
42             cols.add(col); diag1.add(row-col); diag2.add(row+col)
43             board[row][col] = 'Q'
44             backtrack(row + 1)
45             cols.remove(col); diag1.remove(row-col); diag2.remove(row+col)

```

```

46         board[row][col] = '.'
47     backtrack(0)
48     return result

```

Pruning strategies:

- › **Constraint propagation:** Skip branches that violate constraints immediately
- › **Bound pruning:** Skip branches that cannot improve on best known solution
- › **Symmetry breaking:** Avoid exploring symmetrically equivalent states
- › **Early termination:** Return as soon as one solution found (if only need one)

Backtracking complexity: Exponential in worst case ($O(n!)$, $O(2^n)$), but pruning can dramatically reduce actual runtime.

Backtracking is NOT DP: Backtracking explores all possibilities with pruning; DP stores subproblem results to avoid recomputation. Backtracking problems don't have overlapping subproblems.

3.7 Bit Manipulation

Computers operate on binary; bit tricks exploit this for $O(1)$ operations.

Core Operations

Operation	Expression	Effect
Set bit k	<code>n (1 << k)</code>	$= (1 \ll k)$
Clear bit k	<code>n & ~(1 << k)</code>	Force bit k to 0
Toggle bit k	<code>n ^ (1 << k)</code>	Flip bit k
Check bit k	<code>(n >> k) & 1</code>	Get value of bit k
Clear lowest set bit	<code>n & (n - 1)</code>	Removes rightmost 1 bit
Isolate lowest set bit	<code>n & (-n)</code>	Gets rightmost 1 bit only
Check power of 2	<code>n > 0 and (n & (n-1)) == 0</code>	True iff exactly one bit set
Count set bits	<code>bin(n).count('1')</code> or <code>n.bit_count()</code>	Hamming weight

Key XOR Tricks

```

1  # XOR properties:
2  # a ^ a = 0 (anything XOR itself = 0)
3  # a ^ 0 = a (anything XOR 0 = itself)
4  # XOR is commutative and associative
5
6  # Find single number in array where all others appear twice:
7  def single_number(nums):
8      result = 0
9      for x in nums:
10         result ^= x  # pairs cancel out
11     return result
12
13 # Find two unique numbers when all others appear twice:
14 def single_number_two(nums):
15     xor = 0
16     for x in nums: xor ^= x  # xor of the two unique numbers
17     # Find any set bit in xor (they differ in this bit)

```

```

18     diff_bit = xor & (-xor)
19     a = 0
20     for x in nums:
21         if x & diff_bit:
22             a ^= x # group by this bit
23     return a, xor ^ a
24
25 # Swap without temp:
26 a ^= b; b ^= a; a ^= b
27
28 # Check if two integers have opposite signs:
29 def opposite_signs(a, b):
30     return (a ^ b) < 0 # MSB will be 1 if signs differ

```

Bitmask DP

```

1 # Enumerate all subsets of a bitmask:
2 mask = 0b1011 # some bitmask
3 sub = mask
4 while sub > 0:
5     process(sub) # sub is a subset of mask
6     sub = (sub - 1) & mask # get next subset
7
8 # Check if bit i is set in mask:
9 if mask & (1 << i):
10     pass
11
12 # Count set bits (Brian Kernighan's algorithm):
13 def count_bits(n):
14     count = 0
15     while n:
16         n &= (n - 1) # remove lowest set bit
17         count += 1
18     return count
19
20 # Generate all 2^n subsets:
21 n = 4
22 for mask in range(1 << n):
23     subset = [i for i in range(n) if mask & (1 << i)]

```

Bit Tricks Summary

```

1 # Multiply/divide by power of 2:
2 x << k == x * 2^k
3 x >> k == x // 2^k
4
5 # Modulo by power of 2:
6 x & (m - 1) == x % m # only works when m is power of 2
7
8 # Average without overflow:
9 avg = (a & b) + ((a ^ b) >> 1)
10
11 # Absolute value (no branching):
12 mask = x >> 31 # -1 if negative, 0 if positive
13 abs_x = (x + mask) ^ mask

```

```

14
15 # Next power of 2:
16 def next_power_of_2(n):
17     n -= 1
18     n |= n >> 1; n |= n >> 2; n |= n >> 4
19     n |= n >> 8; n |= n >> 16
20     return n + 1

```

3.8 Number Theory / Math

GCD and LCM

```

1 import math
2
3 # GCD using Euclidean algorithm: gcd(a, b) = gcd(b, a % b)
4 def gcd(a, b):
5     while b:
6         a, b = b, a % b
7     return a
8
9 # LCM: lcm(a, b) = a * b / gcd(a, b)
10 def lcm(a, b):
11     return a * b // gcd(a, b)
12
13 # Python 3.9+:
14 math.gcd(a, b)
15 math.lcm(a, b)
16
17 # GCD complexity: O(log(min(a, b))) - Fibonacci numbers are worst case

```

Extended Euclidean Algorithm: Finds x, y such that $a \cdot x + b \cdot y = \gcd(a, b)$:

```

1 def extended_gcd(a, b):
2     if b == 0:
3         return a, 1, 0
4     g, x, y = extended_gcd(b, a % b)
5     return g, y, x - (a // b) * y

```

Sieve of Eratosthenes

Find all primes up to n in $O(n \log \log n)$:

```

1 def sieve(n):
2     is_prime = [True] * (n + 1)
3     is_prime[0] = is_prime[1] = False
4     for i in range(2, int(n**0.5) + 1):
5         if is_prime[i]:
6             for j in range(i*i, n+1, i): # start from i^2 (smaller multiples already marked)
7                 is_prime[j] = False
8     return [i for i in range(n+1) if is_prime[i]]
9
10 # Segmented sieve for large ranges:
11 # Prime factorization using sieve (smallest prime factor):

```

```

12 def smallest_prime_factor(n):
13     spf = list(range(n + 1))
14     for i in range(2, int(n**0.5) + 1):
15         if spf[i] == i: # i is prime
16             for j in range(i*i, n+1, i):
17                 if spf[j] == j:
18                     spf[j] = i
19     return spf

```

Modular Arithmetic

For large number problems where answer is mod $10^9 + 7$:

```

1  MOD = 10**9 + 7
2
3  # Basic operations:
4  (a + b) % MOD
5  (a - b + MOD) % MOD # add MOD before % to avoid negative
6  (a * b) % MOD
7  (a ** b) % MOD == pow(a, b, MOD) # Python's built-in is efficient
8
9  # Modular inverse (when MOD is prime, use Fermat's little theorem):
10 # a^(MOD-1) ≡ 1 (mod MOD) when MOD is prime
11 # Therefore a^(-1) ≡ a^(MOD-2) (mod MOD)
12 def mod_inverse(a, mod):
13     return pow(a, mod - 2, mod) # O(log mod)
14
15 # Modular inverse using extended GCD (works for any mod, gcd(a,mod)=1):
16 def mod_inverse_gcd(a, mod):
17     g, x, _ = extended_gcd(a, mod)
18     if g != 1: raise ValueError("Inverse doesn't exist")
19     return x % mod
20
21 # Modular combinations:
22 def mod_choose(n, k, mod=10**9+7):
23     if k > n: return 0
24     # Precompute factorials
25     fact = [1] * (n + 1)
26     for i in range(1, n+1):
27         fact[i] = fact[i-1] * i % mod
28     inv_fact = [1] * (n + 1)
29     inv_fact[n] = pow(fact[n], mod - 2, mod)
30     for i in range(n-1, -1, -1):
31         inv_fact[i] = inv_fact[i+1] * (i+1) % mod
32     return fact[n] * inv_fact[k] % mod * inv_fact[n-k] % mod

```

Fast Exponentiation (Binary Exponentiation)

Compute a^b in $O(\log b)$:

```

1  def fast_pow(base, exp, mod=None):
2      result = 1
3      while exp > 0:
4          if exp & 1: # if current bit is 1
5              result = result * base
6              if mod: result %= mod

```

```

7     base = base * base
8     if mod: base %= mod
9     exp >>= 1          # shift right (divide by 2)
10    return result
11
12 # Python's built-in pow(a, b, mod) uses this algorithm

```

Key idea: $a^{13} = a^{(1101 \text{ in binary})} = a^8 \cdot a^4 \cdot a^1$. Square the base repeatedly, multiply result when bit is 1.

Prime Factorization

```

1 def prime_factors(n):
2     factors = {}
3     d = 2
4     while d * d ≤ n:
5         while n % d == 0:
6             factors[d] = factors.get(d, 0) + 1
7             n //= d
8         d += 1
9     if n > 1:
10        factors[n] = factors.get(n, 0) + 1
11    return factors
12 # Complexity: O(√n)

```

Useful Math Facts

- › Number of divisors of n : $O(n^{1/3})$ on average
- › Sum of 1 to n : $n(n+1)/2$
- › Sum of squares 1 to n : $n(n+1)(2n+1)/6$
- › Catalan numbers: $C(2n,n)/(n+1)$ — counts BSTs, valid parentheses, polygon triangulations
- › **Pigeonhole principle:** $n+1$ items in n containers $\rightarrow$ at least one container has ≥ 2 items
- › **Stars and bars:** Distribute n identical items into k bins: $C(n+k-1, k-1)$

4 Algorithm Selection Cheat Table

Problem Type	Key Signal	Recommended Algorithm	Complexity
Shortest path, unweighted	BFS in grid/graph	BFS	$O(V+E)$
Shortest path, non-negative weights	Edge weights, single source	Dijkstra	$O((V+E)\log V)$
Shortest path, negative weights	Negative edges allowed	Bellman-Ford	$O(VE)$
All-pairs shortest path	Need dist between all pairs	Floyd-Warshall	$O(V^3)$
Minimum spanning tree	Connect all nodes cheaply	Kruskal or Prim	$O(E \log E)$
Sort, general	$n \leq 10^6$	Quick sort / Merge sort	$O(n \log n)$

Problem Type	Key Signal	Recommended Algorithm	Complexity
Sort, small integer range	Keys in $[0, k]$	Counting sort / Radix sort	$O(n+k)$
Cycle detection, undirected	Find cycle in graph	Union-Find or DFS	$O(\alpha(n))$ or $O(V+E)$
Cycle detection, directed	Find cycle in digraph	DFS (3-color)	$O(V+E)$
Topological order	DAG dependency order	Kahn's or DFS topo	$O(V+E)$
Connected components	Group nodes	DFS/BFS + visited set	$O(V+E)$
Optimal value with choices	Overlapping subproblems	Dynamic Programming	Varies
Make decisions greedily	Exchange argument holds	Greedy	$O(n \log n)$ or $O(n)$
All subsets/permutations	Enumerate possibilities	Backtracking	$O(2^n)$ or $O(n!)$
Find element in sorted array	Array is sorted	Binary search	$O(\log n)$
Find minimum X satisfying condition	Monotone predicate on integers	Binary search on answer	$O(\log n \cdot \text{check})$
Frequency counting	Count occurrences	Hash map / Counting array	$O(n)$
Range minimum/maximum	Static range queries	Sparse table	$O(n \log n)$ pre, $O(1)$ query
Range sum queries	Prefix sums or updates	Prefix sum / Fenwick tree	$O(n)$ or $O(n \log n)$
Substring search	Pattern in text	KMP / Rabin-Karp	$O(n+m)$
Disjoint set operations	Dynamic connectivity	Union-Find	$O(\alpha(n))$ amortized
Large integers, divisibility	Modular arithmetic	Fermat's theorem, sieve	$O(\log n)$
Find unique / pair in XOR	XOR properties	Bit manipulation	$O(n)$
Multiple choices at each step	Combinatorial search	Backtracking + pruning	Exponential
Maximize non-overlapping intervals	Interval scheduling	Greedy (sort by end time)	$O(n \log n)$

5 Real-Life Analogies

Algorithm / Concept	Real-Life Analogy
BFS	Ripple in a pond — spreads layer by layer from center
DFS	Solving a maze by always turning left until you hit a wall, then backtracking
Dijkstra	Google Maps routing — always expand the nearest unvisited city
Bellman-Ford	Send rumors through a network; after $n-1$ rounds, everyone knows the shortest path
Dynamic Programming	Filling out a tax form — use last year's answers to compute this year's
Memoization	Sticky notes — write down each sub-answer once so you never recalculate it

Algorithm / Concept	Real-Life Analogy
Greedy (interval scheduling)	Booking meeting rooms — always pick the meeting that ends soonest
Merge Sort	Library sorting books by splitting into piles, sorting each pile, then merging
Quick Sort	Separating audience by height relative to a pivot person, then recursing
Binary Search	Guessing a number — always guess the middle of the remaining range
Topological Sort	Course prerequisite ordering — take prerequisites before the main course
Union-Find	Friend groups — if A and B are friends, merge their groups; check if same group
Backtracking	Trying all keys on a keyring — discard a key as soon as it doesn't fit
Sieve of Eratosthenes	Crossing out multiples on a number grid to find primes
Amortized Analysis	Buying coffee with a punch card — most coffees are cheap, the 10th is free
Big-O	Speed limit signs — tells you the worst that can happen, not typical speed
MST (Kruskal)	Building cheapest road network by adding roads from cheapest to most expensive
Hash Table	Lockers with assigned numbers — direct access by computing the locker number
Heap	Hospital triage — the most critical patient always treated next

6 Common Interview Questions

6.1 Complexity Analysis

- › What is the time complexity of Python's `list.sort()`? ($O(n \log n)$ — Tim Sort)
- › What is the amortized complexity of appending to a Python list? ($O(1)$)
- › Given a recursive function, derive its time complexity using Master Theorem
- › What is the space complexity of BFS vs DFS? (BFS: $O(V)$ for queue; DFS: $O(V)$ for stack/recursion depth)

6.2 Sorting

- › Sort an array of 0s, 1s, and 2s without using built-in sort (Dutch National Flag — $O(n)$)
- › Find the k-th largest element in an array (QuickSelect — $O(n)$ avg, or heap — $O(n \log k)$)
- › Check if an array is sorted in $O(n)$
- › Merge two sorted arrays in-place

6.3 Binary Search

- › Find first and last position of element in sorted array (LeetCode 34)
- › Search in rotated sorted array (LeetCode 33)
- › Find minimum in rotated sorted array (LeetCode 153)
- › Koko eating bananas / minimum capacity ship (binary search on answer space)
- › Find peak element (LeetCode 162 — binary search on unsorted array!)

6.4 Graphs

- › Number of islands (DFS/BFS on grid — LeetCode 200)
- › Course schedule — detect cycle in directed graph (LeetCode 207)
- › Word ladder — BFS shortest path (LeetCode 127)
- › Clone graph (DFS with memo)
- › Cheapest flights within k stops (modified Dijkstra/Bellman-Ford)
- › Reconstruct itinerary (Eulerian path via DFS)
- › Network delay time (Dijkstra — LeetCode 743)
- › Find if path exists — Union-Find
- › Minimum cost to connect all points — Prim's MST (LeetCode 1584)
- › Critical connections / bridges (Tarjan's algorithm)

6.5 Dynamic Programming

- › Climbing stairs / coin change (basic 1D DP)
- › Longest common subsequence (classic 2D DP)
- › Knapsack variants (0/1, unbounded, multi-dimensional)
- › Longest increasing subsequence ($O(n^2)$ DP or $O(n \log n)$ patience sort)
- › Edit distance / regular expression matching
- › Maximum subarray sum (Kadane's algorithm — $O(n)$ DP)
- › Maximum product subarray
- › House robber / house robber II (circular)
- › Unique paths / minimum path sum in grid
- › Burst balloons (interval DP — LeetCode 312)
- › Word break (1D DP + hash set)
- › Partition equal subset sum (subset sum DP)

6.6 Backtracking

- › N-Queens (LeetCode 51)
- › Sudoku solver (LeetCode 37)
- › Generate all valid parentheses (LeetCode 22)
- › Subsets / permutations / combinations (LeetCode 78, 46, 77)
- › Word search in grid (LeetCode 79)

6.7 Greedy

- › Merge overlapping intervals (LeetCode 56)
- › Non-overlapping intervals (LeetCode 435)
- › Jump game I and II (LeetCode 55, 45)
- › Task scheduler (LeetCode 621)
- › Minimum number of arrows to burst balloons (LeetCode 452)

7 Typical Mistakes Candidates Make

7.1 Complexity Analysis Mistakes

- › **Forgetting recursion call stack space** — $O(n)$ space for $O(n)$ deep recursion
- › **Counting operations instead of asymptotic complexity** — writing " $O(2n)$ " instead of " $O(n)$ "
- › **Wrong space complexity for BFS** — BFS can hold $O(V)$ nodes in queue, not $O(1)$
- › **Not accounting for sorting** — saying a solution is $O(n)$ when it first sorts in $O(n \log n)$

- › **Misapplying Master Theorem** — forgetting to check regularity condition for Case 3

7.2 Graph Algorithm Mistakes

- › **Using Dijkstra with negative edges** — it will produce wrong results
- › **Not handling disconnected graphs** — BFS/DFS from single source misses other components
- › **Off-by-one in Bellman-Ford** — need exactly $V-1$ iterations, not V
- › **Forgetting to check for cycles in topological sort** — not verifying all nodes are processed
- › **Confusing directed and undirected cycle detection** — using undirected method on directed graph
- › **Not initializing all distances to infinity** — leaving 0 causes wrong Dijkstra results

7.3 Sorting Mistakes

- › **Using comparison sort when counting/radix would be $O(n)$**
- › **Forgetting stability requirements** — using unstable sort when stability matters
- › **Not randomizing quicksort pivot** — gets $O(n^2)$ on sorted input in interviews where test cases include sorted arrays
- › **Forgetting in-place constraint** — using merge sort when $O(1)$ space needed

7.4 Dynamic Programming Mistakes

- › **Defining state incorrectly** — most DP bugs come from wrong state definition
- › **Wrong iteration order** — 1D knapsack must iterate weights in reverse for 0/1; forward for unbounded
- › **Off-by-one in indices** — $dp[i]$ represents item i or first i items?
- › **Not thinking about base cases** — wrong base cases cause cascade of wrong answers
- › **Confusing DP with backtracking** — implementing exponential backtracking when DP is needed
- › **Forgetting to return correct cell** — computing $dp[n][m]$ when answer is in $dp[n-1][m-1]$

7.5 Binary Search Mistakes

- › **Integer overflow with $mid = (lo + hi) / 2$** — use $lo + (hi - lo) // 2$
- › **Infinite loop** — wrong loop condition or mid not shrinking the range
- › **Off-by-one in boundary** — $lo < hi$ vs $lo \leq hi$; $hi = mid$ vs $hi = mid - 1$
- › **Wrong convergence** — `lower_bound` needs $hi = len(arr)$, not $len(arr) - 1$

7.6 General Interview Mistakes

- › **Not clarifying constraints** — solution for $n=10$ differs from $n=10^9$
- › **Not discussing complexity** — implementing a solution without analyzing it
- › **Jumping to code** — not spending time designing the algorithm first
- › **Not testing with edge cases** — empty input, single element, all duplicates
- › **Over-engineering** — writing complex code when a simple $O(n^2)$ is acceptable for given n

8 Revision Cheat Sheet

8.1 Must-Know Complexities

Operation	Data Structure	Time
Access by index	Array	$O(1)$
Search	Array (unsorted)	$O(n)$
Search	Array (sorted)	$O(\log n)$
Insert/Delete	Array (arbitrary position)	$O(n)$
Insert/Delete	Linked List	$O(1)$ at head; $O(n)$ to find
Push/Pop	Stack/Queue	$O(1)$
Insert/Delete/Search	Hash Table	$O(1)$ avg, $O(n)$ worst
Insert/Delete/Search	BST (balanced)	$O(\log n)$
Insert/Extract-Min	Min-Heap	$O(\log n)$
Build heap from array	Heap	$O(n)$
DFS/BFS	Graph	$O(V+E)$
Dijkstra	Graph (non-neg)	$O((V+E) \log V)$
Bellman-Ford	Graph	$O(VE)$
Floyd-Warshall	Graph	$O(V^3)$
Kruskal/Prim	Graph	$O(E \log E)$
Merge Sort / Heap Sort	Array	$O(n \log n)$
Quick Sort	Array	$O(n \log n)$ avg
Counting Sort	Array	$O(n+k)$
Binary Search	Sorted Array	$O(\log n)$
Union-Find	DSU	$O(\alpha(n))$ amortized
Sieve of Eratosthenes	-	$O(n \log \log n)$
GCD	-	$O(\log(\min(a,b)))$
Fast Exponentiation	-	$O(\log n)$

8.2 Algorithm Triggers (What to Think When You See...)

You see...	Think...
Shortest path, no weights	BFS
Shortest path, positive weights	Dijkstra
Shortest path, negative weights	Bellman-Ford
All-pairs shortest path	Floyd-Warshall
Minimum spanning tree	Kruskal (sparse) / Prim (dense)
"Connected components" or "groups"	Union-Find or DFS
"Cycle in directed graph"	Topological sort (Kahn's) or DFS 3-color
"Dependency ordering"	Topological sort
"Count ways", "min/max cost"	Dynamic Programming
"Optimal choice at each step"	Greedy (verify with exchange argument)
"All possibilities / enumerate"	Backtracking
"Sorted array, find X"	Binary search
"Find minimum X such that..."	Binary search on answer
"Sliding window, subarray"	Two pointers or sliding window
"Top K elements"	Heap (size K)
"Anagram / frequency"	Hash map or counting array

You see...	Think...
"Intervals, overlapping"	Sort by start/end + greedy
"Parentheses, matching"	Stack
"Next greater element"	Monotonic stack
"Integer keys, small range"	Counting sort / bucket
"Large integers mod prime"	Modular arithmetic + Fermat
"Find single number in pairs"	XOR trick
"Power of 2 / bit patterns"	Bit manipulation
"String matching"	KMP / Rabin-Karp
"2D grid, path finding"	BFS (shortest) / DFS (any path)
"Subsets, combinations"	Backtracking or bitmask
"Repeated subproblems visible"	Memoize it → DP

8.3 Golden Rules

1. **Always clarify constraints before coding** — n determines your algorithm
2. **State complexity before implementation** — shows algorithmic thinking
3. **Draw examples** — small examples reveal patterns
4. **Code the brute force first if stuck** — then optimize
5. **Test edge cases explicitly** — empty, single element, all same, already sorted
6. **For graphs:** always check — directed or undirected? weighted? negative edges? disconnected?
7. **For DP:** define state precisely, write recurrence, then code
8. **For greedy:** justify why local optimal = global optimal
9. **Binary search:** maintain invariant, never drop the answer
10. **Recursion:** trust the recursion — define what it returns, use it correctly